

TMUA EUCLID

Sample Paper 2

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Introduction for students

These solutions are written to show how a successful approach can be built one step at a time. The explanations are deliberately fuller than anything you would need to write in the TMUA itself, which is a multiple-choice test.

Try each question before reading on. When you check a solution, focus first on the opening idea: recognising a useful identity, choosing a substitution, or turning the wording into an equation. Those decisions are often more important than the final calculation.

There may be other correct methods. The method shown is chosen because it is reasonably short while still making the important reasoning visible.

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Question 1

The chain must begin with a known truth and finish at the required inequality. Start with

$$(a - b)^2 \geq 0.$$

Expanding and rearranging gives

$$a^2 + b^2 \geq 2ab.$$

Adding $2ab$ to both sides,

$$(a + b)^2 \geq 4ab.$$

Since $a + b = 1$, this becomes $1 \geq 4ab$. Both variables are positive, so $ab > 0$ and division does not reverse the inequality:

$$\frac{1}{ab} \geq 4.$$

Finally,

$$\frac{1}{a} + \frac{1}{b} = \frac{a + b}{ab} = \frac{1}{ab}.$$

Combining the last two lines proves the result. The order is P, Q, R, S, T, U .

Answer: option A.

Question 2

The recurrence is second order: computing a_{k+1} requires both a_k and a_{k-1} . However, the induction hypothesis assumes only that a_k is even. When the proposed proof writes

$$a_{k+1} = a_k + a_{k-1},$$

it has not established that a_{k-1} is even.

The underlying problem is that only a_1 is specified. The value of a_2 is unconstrained, whereas a second-order recurrence needs two consecutive starting values. The claim is in fact false: choose

$$a_1 = 2, \quad a_2 = 1.$$

The sequence begins

$$2, 1, 3, 4, 7, 11, \dots,$$

so a_1 is even but the later terms are not all even.

Answer: option B.

Question 3

Write out the first terms:

$$a_2 = 1 - a_1, \quad a_3 = 1 - a_2 = 1 - (1 - a_1) = a_1.$$

Thus the sequence repeats after two steps for every starting value. Its least period is therefore either 1 or 2.

The least period is 1 exactly when the sequence is constant, which requires

$$1 - a_1 = a_1 \iff a_1 = \frac{1}{2}.$$

Hence the least period is 2 exactly when $a_1 \neq 1/2$. This is precisely statement Q , so P and Q are equivalent.

Answer: option A.

Question 4

Negate one layer at a time. The original statement has the form

for every real x , there exists a positive integer n such that $x < n \leq x + 2$.

Negating reverses both quantifiers. The condition $x < n \leq x + 2$ is a conjunction, so its negation is

$$n \leq x \quad \text{or} \quad n > x + 2.$$

Therefore the negation is: there exists a real x such that every positive integer n satisfies one of these two inequalities.

Answer: option C.

Question 5

If

$$f(0)f(1) < 0,$$

then the endpoint values have opposite signs. Because f is continuous, its graph must cross the x -axis somewhere strictly between 0 and 1. Thus $Q \Rightarrow P$.

The reverse implication is false. For example,

$$f(x) = \left(x - \frac{1}{2}\right)^2$$

is continuous and has a zero at $x = 1/2$, so P holds. But

$$f(0)f(1) = \frac{1}{4} \cdot \frac{1}{4} > 0,$$

so Q fails. Hence P is necessary but not sufficient for Q .

Answer: option C.

Question 6

For statement I, substitute $n = 10$:

$$n^2 + n + 11 = 100 + 10 + 11 = 121 = 11^2,$$

which is composite.

For statement II, if $n \equiv 0 \pmod{11}$, then $11 \mid n(n+1)$. If $n \equiv 10 \pmod{11}$, then $11 \mid n+1$, so again $11 \mid n(n+1)$. Thus in either case

$$11 \mid n^2 + n + 11.$$

For positive n , this value is greater than 11, so it is composite.

Statement III is false. The value $n = 14$ has remainder 3 modulo 11, but

$$14^2 + 14 + 11 = 221 = 13 \cdot 17.$$

Answer: option E.

Question 7

Let $a_n = a$ and $a_{n+1} = b$. Repeated use of $a_{n+2} = a_{n+1} - a_n$ gives

$$\begin{aligned}a_{n+2} &= b - a, & a_{n+3} &= -a, \\a_{n+4} &= -b, & a_{n+5} &= a - b, \\a_{n+6} &= a.\end{aligned}$$

Therefore

$$a_{n+3} = -a_n \quad \text{and} \quad a_{n+6} = a_n,$$

so statements I and II are true. Also,

$$a + b + (b - a) - a - b + (a - b) = 0,$$

which proves statement III. Since a and b were arbitrary, all three statements hold.

Answer: option H.

Question 8

Write

$$24 = 2^3 \cdot 3.$$

If $24 \mid n^2$, then $8 \mid n^2$. The exponent of 2 in a square is even, so it must actually be at least 4; hence $2 \mid n$. Also, $3 \mid n^2$ implies $3 \mid n$. Therefore $6 \mid n$, so $P \Rightarrow Q$.

The reverse implication fails for $n = 6$: clearly $6 \mid n$, but $n^2 = 36$ is not divisible by 24. Thus P is sufficient but not necessary for Q .

Answer: option B.

Question 9

Since W is the midpoint of UV ,

$$WU = WV = \frac{1}{2}UV.$$

Thus $WT = UV/2$ is equivalent to $WT = WU = WV$, proving the equivalence of Q and R .

If $\angle UTV = 90^\circ$, then T lies on the circle with diameter UV . Its centre is W and its radius is $UV/2$, so $WT = UV/2$. Conversely, if $WT = UV/2$, then T lies on this circle, and the angle in a semicircle is a right angle. Hence P and Q are equivalent.

Therefore P , Q , and R are mutually equivalent, and all three statements are true.

Answer: option G.

Question 10

Using

$$p = a + b - c, \quad q = a + c - b, \quad r = b + c - a,$$

we find

$$q - p = 2(c - b) > 0, \quad r - q = 2(b - a) > 0.$$

Thus $p < q < r$, proving statement I.

Statement II is false. If $a = 2$, $b = 3$, and $c = 4$, then the original sides form a triangle, but

$$p = 1, \quad q = 3, \quad r = 5,$$

and $p + q < r$.

Finally, adding the three definitions gives

$$p + q + r = a + b + c,$$

so statement III is true.

Answer: option F.

Question 11

Factorise:

$$n^2 + 3n + 2 = (n + 1)(n + 2).$$

For divisibility by 5, we require $n \equiv 3$ or $4 \pmod{5}$. For divisibility by 7, we require $n \equiv 5$ or $6 \pmod{7}$. Combining the choices gives

$$n \equiv 13, 19, 33, 34 \pmod{35}.$$

There are 14 complete blocks of 35 from 1 to 490, each containing four valid values. This gives

$$14 \cdot 4 = 56.$$

The remaining integers 491 to 500 have residues 1 through 10, so none is valid. The total is 56.

Answer: option C.

Question 12

Assume a smallest positive-integer solution (m, n) exists. From

$$m^2 = 2n^2,$$

the number m is even, so write $m = 2p$. Substitution gives

$$n^2 = 2p^2,$$

so n is also even; write $n = 2q$. Substituting again gives

$$p^2 = 2q^2.$$

Thus (p, q) is another positive-integer solution, but $p = m/2 < m$, contradicting the minimality of m . The correct order is P, Q, R, S, T, U .

Answer: option A.

Question 13

If f has four distinct roots $x_1 < x_2 < x_3 < x_4$, then between each pair of consecutive roots the graph must have a stationary point. Therefore f' has a root in each of

$$(x_1, x_2), \quad (x_2, x_3), \quad (x_3, x_4).$$

Thus statement I is true. Applying the same graph argument to three distinct roots of f' gives two distinct roots of f'' , so statement II is true.

The converse is false. Take

$$f(x) = x^4 - x^2 + 1.$$

Then $f''(x) = 12x^2 - 2$ has the two distinct roots $x = \pm 1/\sqrt{6}$, but f has no real roots: treating it as a quadratic in x^2 gives discriminant $1 - 4 = -3$. Hence statement III is false.

Answer: option D.

Question 14

Integrating and using $f(0) = 0$ gives

$$f(x) = \frac{x^3}{3} - 4x.$$

The stationary points are $x = \pm 2$, with

$$f(-2) = \frac{16}{3}, \quad f(2) = -\frac{16}{3}.$$

Therefore $f(x) = k$ has three real solutions exactly when $|k| < 16/3$, proving statement I.

The equation is $x^3 - 12x - 3k = 0$. Comparing coefficients in $(x - \alpha)(x - \beta)(x - \gamma)$ gives

$$\alpha + \beta + \gamma = 0, \quad \alpha\beta + \beta\gamma + \gamma\alpha = -12.$$

Hence $\alpha^2 + \beta^2 + \gamma^2 = 24$, so statement II is true.

For $0 < k < 16/3$, the middle crossing lies to the left of 0, because the cubic is decreasing through the origin. Thus the middle root is negative, so statement III is false.

Answer: option E.

Question 15

The divisibility condition

$$24 \mid (m^2 - 1)$$

holds exactly when m is odd and not divisible by 3. Indeed, odd m gives $8 \mid m^2 - 1$, while $3 \mid m^2 - 1$ exactly when $3 \nmid m$.

There are 50 odd integers from 1 to 100. The odd multiples of 3 are

$$3, 9, 15, \dots, 99,$$

of which there are 17. Therefore exactly

$$50 - 17 = 33$$

integers satisfy the condition.

The statement S_n says that at least n integers satisfy it. Since exactly 33 do, S_n is true exactly for $n \leq 33$. Thus precisely $S_1, S_2, \dots, S_{33}$ are true.

Answer: option C.

Question 16

The function f is not wobbly, so $f'(x) \neq 0$ for every real x .

For $g(x) = f(x)^2$,

$$g'(x) = 2f(x)f'(x).$$

Since $f(-1) < 0 < f(1)$, continuity gives a point $c \in (-1, 1)$ with $f(c) = 0$. Hence $g'(c) = 0$, proving statement I.

For

$$h(x) = f(x^2 - 2x) + f(x^2 + 2x),$$

the chain rule gives

$$h'(0) = -2f'(0) + 2f'(0) = 0,$$

so statement II is true.

Finally, if $p(x) = f(x^3 + x)$, then

$$p'(x) = f'(x^3 + x)(3x^2 + 1).$$

Both factors are non-zero, so p' never vanishes. Statement III is false.

Answer: option E.

Question 17

Complete the square:

$$f(x) = x^2 - 2x = (x - 1)^2 - 1.$$

Put $y = f(x)$. Then $f(y) = k$ gives

$$y = 1 \pm \sqrt{k + 1}.$$

Each value of y produces two distinct x -values exactly when $y > -1$. Both values satisfy this precisely when

$$-1 < k < 3,$$

proving statement I.

Expanding $f(f(x)) = k$ gives

$$x^4 - 4x^3 + 2x^2 + 4x - k = 0.$$

The product of the four roots is the constant term $-k$, not k , so statement II is false.

Within the four-root range, the larger value of y gives two roots outside $(0, 2)$. The smaller value gives two roots in $(0, 2)$ exactly when it is negative:

$$1 - \sqrt{k + 1} < 0 \quad \Longleftrightarrow \quad k > 0.$$

Thus statement III is true.

Answer: option F.

Question 18

The derivative

$$f'(x) = (x^2 - 1)(x^2 - 4)$$

is even. For $F(x) = f(x) + f(-x)$,

$$F'(x) = f'(x) - f'(-x) = 0.$$

Thus F is constant, and $F(0) = 2f(0)$, proving statement I.

A sign chart at $-2, -1, 1, 2$ shows that f' changes from negative to positive at -1 and 2 . These are exactly the two local minima, so statement II is true.

Integrating gives

$$f(x) = \frac{x^5}{5} - \frac{5x^3}{3} + 4x + C.$$

At the four stationary points,

$$f(-1) = -\frac{38}{15} + C, \quad f(-2) = -\frac{16}{15} + C, \quad f(2) = \frac{16}{15} + C, \quad f(1) = \frac{38}{15} + C.$$

Therefore $f(-1) < f(-2) < f(2) < f(1)$, proving statement III.

Answer: option H.

Question 19

Put

$$t = |x - 2| \geq 0.$$

The equation becomes

$$|t^2 - 1| = kt + \frac{3}{2}.$$

Since $t = 0$ is not a solution, six distinct values of x correspond to exactly three distinct positive values of t .

For $0 \leq t \leq 1$, the equation is

$$t^2 + kt + \frac{1}{2} = 0.$$

It has two distinct roots in $(0, 1)$ exactly when

$$k^2 > 2, \quad k < 0, \quad 1 + k + \frac{1}{2} > 0.$$

Together these give

$$-\frac{3}{2} < k < -\sqrt{2}.$$

For $t \geq 1$, the equation is

$$t^2 - kt - \frac{5}{2} = 0.$$

Its roots have negative product, so exactly one is positive; throughout the same interval of k , that positive root is greater than 1. Hence there are three positive t -values and six x -values. At either endpoint, roots merge or overlap, so the inequalities are strict.

Answer: option B.

Question 20

For statement P , regard the equation as a quadratic in z :

$$az^2 + 2xz + (x^2 - 1) = 0.$$

The case $a = 0$ fails when $x = 0$. For $a \neq 0$, a real z exists for every x exactly when

$$(1 - a)x^2 + a \geq 0$$

for all real x . This is equivalent to

$$0 < a \leq 1.$$

Thus $P \iff 0 < a \leq 1$.

For statement Q , regard the equation as a quadratic in x :

$$x^2 + 2zx + (az^2 - 1) = 0.$$

A real x exists for every z exactly when

$$(1 - a)z^2 + 1 \geq 0$$

for all real z , which is equivalent to $a \leq 1$. Hence $Q \iff a \leq 1$, while $R \iff a > 0$.

Therefore P holds exactly when both Q and R hold. If Q holds, then either $a > 0$ and P holds, or $a \leq 0$ and R fails. Similarly, if R holds, then either $a \leq 1$ and P holds, or $a > 1$ and Q fails. All three statements are true.

Answer: option G.